

MAT415 Lecture Notes

Arky!! :3c

'26 Fall Semester

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Day 1: Gaussian Integers (Sep. 8, 2026)

The Quercus site for this class will be up tonight or early tomorrow. This class will have (approximately) weekly homework. There will be regular oral quizzes with the TA, about 5–10 minutes, where you will discuss one of the problems on the homework.

Consider the following example,

$$1 + 1 = 2, \quad 1 + 4 = 5, \quad 4 + 9 = 13, \quad 1 + 16 = 17.$$

Theorem 1.1 (Fermat). Let $p > 2$; p can be expressed as a sum of two squares if and only if $p \equiv 1 \pmod{4}$.

Proof. We start with the easy direction. Observe that the squares of the numbers $0, 1, 2, 3, \dots$ are all equal to 0 or 1 modulo 4, i.e., $0^2 = 0$, $1^2 = 1$, $2^2 = 4 \equiv 0 \pmod{4}$, and $3^2 = 9 \equiv 1 \pmod{4}$, and so on (you can always express a number as either $2n$ or $2n + 1$, then squaring, to check that this pattern continues). It follows that if $p = n^2 + m^2$, then $p \equiv 1 \pmod{4}$; if it were the sum of the squares of two odd numbers, then p would be even, which contradicts it being prime.

For the other direction, consider the ring of Gaussian integers,

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\} \subset \mathbb{C}.$$

Consider $p = n^2 + m^2$. Notice that the right hand side cannot be factored as *integers*, but it could be factored in $\mathbb{Z}[i]$ as $(n + im)(n - im)$; in this manner, we may instead ask the question of how to factor $p \in \mathbb{Z}[i]$. The continuation of this proof will use the fact that $\mathbb{Z}[i]$ is a UFD, but we will not prove that today. In particular, $\mathbb{Z}[i]$ is a Euclidean domain with $\alpha \mapsto |\alpha|^2$ is a Euclidean norm. $\square$

Given $z = x + iy \in \mathbb{Z}[i]$, define the norm of z as $N(z) = z \cdot \bar{z} = x^2 + y^2 \in \mathbb{Z}_{\geq 0}$.¹ Observe that we have $N(z \cdot w) = N(z) \cdot N(w)$.

Claim 1.2. $z \in \mathbb{Z}[i]$ is a unit if and only if $N(z) = 1$, i.e., the units are $\{\pm 1, \pm i\}$.

Proof. Suppose $N(z) = 1$; by definition, we see $z \cdot \bar{z} = 1$, and so $\bar{z}$ constitutes the inverse of z . Conversely, suppose $z \in \mathbb{Z}[i]$ is a unit; if there exists another number w such that $z \cdot w = 1$, then, since norm is multiplicative, we may take the norm of both sides to get that $N(z) \cdot N(w) = 1$. Since $N(z) \in \mathbb{Z}_{\geq 0}$, it must be that $N(z) = 1$. $\square$

Claim 1.3. If p is a prime such that $p \equiv 1 \pmod{4}$, then p is not prime in $\mathbb{Z}[i]$.

We will prove the claim later; for now, observe that 1.1 and 1.3 imply each other; one is obvious, and for the direction that the claim implies the theorem, suppose p is not prime, so we may write it as the product of two non-units, $p = \alpha \cdot \beta$, where $\alpha, \beta \in \mathbb{Z}[i]$. We have that $p^2 = N(p) = N(\alpha) \cdot N(\beta)$; but $N(\alpha), N(\beta) \neq 1$, so it must be that $N(\alpha) = N(\beta) = p$. By writing $\alpha = a + bi$, we get that $N(\alpha) = a^2 + b^2 = p$, which is the desired expression as seen in 1.1.

We now prove 1.3 in two ways. For the first way, consider Wilson's theorem:

Theorem 1.4 (Wilson). Let p be prime; then $(p - 1)! \equiv -1 \pmod{p}$.

Proof. The proof of this is simple; since $\mathbb{Z}/p$ is a field, every nonzero residue modulo p has an inverse, where $\alpha \equiv \alpha^{-1}$ if and only if $\alpha \equiv 1$ or -1 , i.e., $\alpha^2 = 1$. It follows that the product of $1, \dots, p - 1$ without 1 and $p - 1$ is 1, so we are done. $\square$

¹at this point, Rozenblyum made a comment about complex number norms versus algebra norms. idk. sure.

Corollary 1.5. If $p = 4n + 1$, then $((2n)!)^2 \equiv -1 \pmod{p}$.

Proof. By Wilson's theorem, we have that $-1 \equiv (p-1)!$. Let us write the factorial out;

$$-1 \equiv (p-1)! = (1 \cdot 2 \cdot \dots \cdot 2n)((p-2n)(p-2n+1) \dots (p-1)).$$

Since we're working modulo p , we may realize the right hand side expression as

$$\dots \equiv (1 \cdot 2 \cdot \dots \cdot 2n)((-2n)(-2n+1) \dots (-1)) = (2n)!(-1)^{2n}(2n)! = ((2n)!)^2. \quad \square$$

The above corollary allows us to write²

$$p \mid ((2n)!)^2 + 1,$$

which we may factor as $((2n)! + i)((2n)! - i)$; if p was prime, then it divides one of them, but we know that $p \mid (a+bi) = (2n)! \pm i$ can't happen.

We now proceed to the second proof. Consider the cyclic group

$$(\mathbb{Z}/p)^\times = \mathbb{Z}/(p-1);$$

if $p = 4n + 1$, then $\mathbb{Z}/(p-1) \cong \mathbb{Z}/(4n)$, whence the existence of some element of order 4 in $(\mathbb{Z}/p)^\times$. Let such an element be called m ; it follows that $m^4 \equiv 1 \pmod{p}$ and $m^2 \equiv -1 \pmod{p}$, but then we would have $p \mid m^2 + 1$, which is a contradiction. While this proof was fast, it didn't really tell us much about the arithmetic that went on in the first proof.

The morals as to the two above proofs are that

- (i) It is important to understand arithmetic in more general number rings (such as prime factorization). As an example,

$$\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$$

is not a UFD (see the MAT347 notes for a reference on this).

- (ii) Questions about number rings are closely related to questions in modular arithmetic.

²Rozenblyum proceeded to stand stunned in front of the blackboard for 3 minutes. When pointed out on how to continue the discussion, he said that this was proof your "IQ drops precipitously when in front of a classroom".

Day 2: Algebra Preliminaries; Algebraic Numbers and Integers (Sep. 10, 2026)

We set some conventions for this class. Unless specified otherwise, a ring will always refer to a commutative ring with unity.

Definition 2.1 (Preliminaries). Let R be a ring.

- (i) R is an *integral domain* if it admits no zero divisors, i.e., whenever $ab = 0$ for $a, b \in R$, then $a = 0$ or $b = 0$.
- (ii) We will write $a \mid b$ if “ a divides b ” for $a, b \in R$ if there exists some $c \in R$ such that $b = ac$.
- (iii) An integral domain R is a *principal ideal domain* (PID) if every ideal is principal, i.e., generated by a single element. Specifically, if $I \subset R$ is an ideal, then there exists $a \in R$ such that $I = \langle a \rangle$.
- (iv) An element $u \in R$ is a *unit* if it has a multiplicative inverse. An element $p \in R$ is *irreducible* if it is not a unit, and whenever $p = ab$, either a or b is a unit. An element $p \in R$ is *prime* if, whenever $p \mid ab$, then $p \mid a$ or $p \mid b$.
- (v) An integral domain R is a *unique factorization domain* (UFD) if every non-unit $x \in R$ is uniquely (up to permutation and multiplication by units) a product of irreducible elements.

Proposition 2.2. If R is a UFD, then irreducibles are primes, and primes are irreducible.

Note that primes are always irreducible. You can reference my MAT347 notes for a proof on this.

Proposition 2.3. Let R be a ring. Then $p \in R$ is prime if and only if $R/\langle p \rangle$ is an integral domain.

Proof. Pick $\bar{a}, \bar{b} \in R/\langle p \rangle$ such that $\bar{a}\bar{b} = 0$ in $R/\langle p \rangle$. Then $ab \in \langle p \rangle$, so $p \mid ab$, where $p \mid a$ or $p \mid b$ as p is prime, so $\bar{a} = 0$ or $\bar{b} = 0$. The converse direction is obtained by reversing this proof. $\square$

We now continue writing out the definitions.

Definition 2.4 (Preliminaries, cont.). Let R be a ring.

- (vi) An integral domain R is a *Euclidean domain* (ED) if there exists a map $v: R \setminus \{0\} \rightarrow \mathbb{Z}_{\geq 0}$ such that, for all $a, b \in R$ with $b \neq 0$, there exists $q, r \in R$ such that $a = bq + r$ with $r = 0$ or $v(r) < v(b)$. In addition, for all nonzero $a, b \in R$, v must satisfy $v(a) \leq v(ab)$.

Theorem 2.5. Let R be an integral domain. If R is a Euclidean domain, then it is a PID. If R is a PID, then it is a UFD.

Once again, the proof is in my MAT347 notes. Let us give some examples of this.

- $R = \mathbb{Z}$ is a Euclidean domain with $v(n) = |n|$.
- If F is a field, then $F[x]$ is a Euclidean domain with $v(f(x)) = \deg f$.
- Let n be an integer, and let $R = \mathbb{Z}/n$ be the integers mod n . R is an integral domain if and only if n is prime; in fact, if n is prime, then $\mathbb{Z}/n$ is a field. For a quick proof of this fact, let $x \in \mathbb{Z}$; then $\bar{x} \in \mathbb{Z}/n$ is a unit if and only if x is coprime to n . If p is prime, then $(\mathbb{Z}/p)^\times$ is a group of order $p - 1$.

Theorem 2.6. Let $p \in \mathbb{N}$ be a prime number.

- (i) (Fermat's little theorem) If $a \in \mathbb{Z}$ is coprime to p , then $a^{p-1} \equiv 1 \pmod{p}$.
- (ii) $(\mathbb{Z}/p)^\times$ is a cyclic group.

Corollary 2.7. Let $p \in \mathbb{N}$ be a prime number. Then there exists $g \in \mathbb{Z}$, called the *primitive root modulo* p , such that (i) if a is coprime to p , then $a \equiv g^r \pmod{p}$ for some $r \geq 0$, and (ii) if $r \in \mathbb{Z}$ such that $g^r \equiv 1 \pmod{p}$, then $p - 1 \mid r$.

Definition 2.8. An *algebraic number* α is a complex number $\alpha \in \mathbb{C}$ for which there exists $a_0, \dots, a_{n-1} \in \mathbb{Q}$ (with $n \geq 1$) such that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_1\alpha + a_0 = 0,$$

i.e., $p(\alpha) = 0$ for some monic $p(x) \in \mathbb{Q}[x]$. An *algebraic integer* is a complex number $\alpha \in \mathbb{C}$ such that there exists $a_0, \dots, a_{n-1} \in \mathbb{Z}$ (with $n \geq 1$) such that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_1\alpha + a_0 = 0,$$

i.e., $p(\alpha) = 0$ for some monic $p(x) \in \mathbb{Z}[x]$. Algebraic numbers are denoted by $\mathbb{Q}^{\text{alg}}$ or $\overline{\mathbb{Q}}$, and algebraic integers are denoted by $\mathbb{Z}^{\text{alg}}$ or $\overline{\mathbb{Z}}$. The bar denotes algebraic closure.

In particular, algebraic numbers form a field, and algebraic integers form a ring. This is not obvious from the definition, but we will take it as true for now. Let us give some examples.

- Any $\alpha \in \mathbb{Q}$ is an algebraic number, since we may pick $p(x) = x - \alpha = 0$. Similarly, any $n \in \mathbb{Z}$ is an algebraic integer per the exact same construction.
- $\mathbb{Z}^{\text{alg}} \subset \mathbb{Q}^{\text{alg}}$.
- $\sqrt{-5} \in \mathbb{Z}^{\text{alg}}$, since $(\sqrt{-5})^2 + 5 = 0$.
- $\omega = \frac{1}{2}(-1 + \sqrt{-3})$ an algebraic integer, since $\omega^2 + \omega + 1 = 0$.
- For $n \geq 1$, we have that $\zeta = \exp(2\pi i/n) \in \mathbb{Z}^{\text{alg}}$, since $\zeta^n - 1 = 0$. ζ is the *primitive n^{th} root of unity*.
- Let $\alpha \in \mathbb{Q}^{\text{alg}}$ and $r \in \mathbb{Q}$. Then $r\alpha \in \mathbb{Q}^{\text{alg}}$. We will see later that, for all $\alpha \in \mathbb{Q}^{\text{alg}}$, there exists some $r \in \mathbb{Z}$ such that $r\alpha \in \mathbb{Z}^{\text{alg}}$.
- $\mathbb{Q}^{\text{alg}}$ is countable, since it is the union of the (finitely many) roots of polynomials in $\mathbb{Q}[x]$, of which are countable. $\mathbb{C}$ is not countable, however, so $\mathbb{Q}^{\text{alg}} \subsetneq \mathbb{C}$.
- $e, \pi \notin \mathbb{Q}^{\text{alg}}$.

Another way to characterize $\mathbb{Q}^{\text{alg}}$ and $\mathbb{Z}^{\text{alg}}$ is to consider subsets $V \subset \mathbb{C}$ such that V is a vector space (respectively, a finitely generated abelian group) over $\mathbb{Q}$. As an example, $V = \mathbb{Q}$ is trivially a vector space over $\mathbb{Q}$, and $V = \{r + s\sqrt{-5} \mid r, s \in \mathbb{Q}\}$ is one as well. In particular, it contains the finitely generated abelian group $\{r + s\sqrt{-5} \mid r, s \in \mathbb{Z}\}$.

Lemma 2.9. Let $\alpha \in \mathbb{C}$. Then

- (i) $\alpha \in \mathbb{Q}^{\text{alg}}$ if and only if there exists a nonzero finite dimensional vector space $V \subset \mathbb{C}$ over $\mathbb{Q}$ such that $\alpha V \subset V$.
- (ii) $\alpha \in \mathbb{Z}^{\text{alg}}$ if and only if there exists a finitely generated abelian group $M \subset \mathbb{C}$ such that $\alpha M \subset M$.

Proof. We start by proving (i). In the forwards direction, suppose $\alpha \in \mathbb{Q}^{\text{alg}}$; then $\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_0 = 0$. Let

$$V = \{r_0 + r_1\alpha + \dots + r_{n-1}\alpha^{n-1} \mid r_0, \dots, r_{n-1} \in \mathbb{Q}\},$$

which we readily see as finite-dimensional as a vector space over $\mathbb{Q}$. It follows that $\alpha V \subset V$, since

$$\begin{aligned}\alpha(r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1}) &= r_0\alpha + r_1\alpha^2 + \cdots + r_{n-1}\alpha^n \\ &= r_0\alpha + r_1\alpha^2 + \cdots + r_{n-1}(-a_{n-1}\alpha^{n-1} - \cdots - a_0) \in V.\end{aligned}$$

Conversely, suppose $V \subset \mathbb{C}$ is finite-dimensional with $\alpha V \subset V$. Let $v_1, \dots, v_d$ be a basis of V over $\mathbb{Q}$. For each i , we see that

$$\alpha v_i = \sum_{j=1}^d r_{ij} v_j, \quad r_{ij} \in \mathbb{Q};$$

i.e., $v \mapsto \alpha v$ is represented by a matrix $R := (r_{ij})$ in this basis. Specifically, R is a $d \times d$ matrix with entries in $\mathbb{Q}$, which we may think of as a complex matrix:

$$R \begin{pmatrix} v_1 \\ \vdots \\ v_d \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^d r_{1j} v_j \\ \vdots \\ \sum_{j=1}^d r_{dj} v_j \end{pmatrix} = \alpha \begin{pmatrix} v_1 \\ \vdots \\ v_d \end{pmatrix}.$$

It follows that $\alpha \in \mathbb{C}$ is an eigenvalue of R , whence α is a root of the characteristic polynomial $\det(R - Ix) \in \mathbb{Q}[x]$, so it has to be an algebraic number.

The proof for (ii), i.e., the algebraic integers case, follows analogously. Note that we need to know that the finitely generated abelian group is free; per the classification theorem (of finitely generated abelian groups), every such V can be written in terms of its free part and torsion; but there is no torsion, so we may reiterate our earlier argument as desired. $\square$

Proposition 2.10. $\mathbb{Q}^{\text{alg}} \subset \mathbb{C}$ is a subfield, and $\mathbb{Z}^{\text{alg}} \subset \mathbb{C}$ is a subring.

Proof. Let $\alpha, \beta \in \mathbb{Q}^{\text{alg}}$. Then, per 2.9, there exist nonzero finite-dimensional subspaces $V, W \subset \mathbb{C}$ such that $\alpha V \subset V$ and $\beta W \subset W$. Consider

$$V \cdot W = \{v_1 w_1 + \dots + v_t w_t \mid v_j \in V, w_j \in W\};$$

observing that $v_1, \dots, v_n$ span V and $w_1, \dots, w_m$ span W , we simply have that $v_i w_j$ for $1 \leq i \leq n$ and $1 \leq j \leq m$ span $V \cdot W$. Thus, $V \cdot W$ is finite-dimensional and nonempty.

Now, for any $v \in V$ and $w \in W$, we have that $\alpha \cdot (vw) = (\alpha v) \cdot w \in V \cdot W$, and $\beta \cdot (vw) = v \cdot (\beta w) \in V \cdot W$. By linearity for $z \in V \cdot W$, we have $\alpha z \in V \cdot W$ and $\beta z \in V \cdot W$, whence $(\alpha + \beta)(V \cdot W) \subset V \cdot W$ and $(\alpha \cdot \beta)(V \cdot W) \subset V \cdot W$, so $\alpha + \beta$ and $\alpha\beta$ are both algebraic numbers per 2.9. An analogous argument holds for $\alpha, \beta \in \mathbb{Z}^{\text{alg}}$.

It remains to show that nonzero $\alpha \in \mathbb{Q}^{\text{alg}}$ have inverses, i.e., $\alpha^{-1} \in \mathbb{Q}^{\text{alg}}$. Indeed, the “multiplying by α ” map, $\alpha: V \rightarrow V$, is injective, so it is isomorphic, and so there exists α^{-1} as the inverse map to α . In particular, it preserves V . Alternatively, we may write

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0 \implies \alpha \cdot \left(-\frac{1}{\alpha_0}\right) (\alpha^{n-1} + a_{n-1}\alpha^{n-2} + \cdots + a_1) = 1,$$

which is a more explicit construction for α^{-1} . $\square$

We now look at more examples.

- Every element of $\mathbb{Z}[\sqrt{-5}]$ is an algebraic integer. Indeed, every $\alpha \in \mathbb{Z}[\sqrt{-5}]$ is of the form $a + b\sqrt{-5}$ with $a, b \in \mathbb{Z}$, and there is clearly an easy monic quadratic $p(x) \in \mathbb{Z}[x]$ such that $p(\alpha) = 0$.
- $\alpha = \sqrt{2} + \sqrt{3} \in \mathbb{Z}^{\text{alg}}$, since it satisfies $\alpha^4 - 10\alpha^2 + 1 = 0$.

Proposition 2.11. Let $\alpha \in \mathbb{Q}^{\text{alg}}$. Then there exists a unique monic irreducible polynomial $f(x) \in \mathbb{Q}[x]$ such that $f(\alpha) = 0$. Moreover, if $g(x) \in \mathbb{Q}[x]$ also satisfies $g(\alpha) = 0$, then $f \mid g$.

Proof. Consider $I = \{h(x) \in \mathbb{Q}[x] \mid h(\alpha) = 0\} \neq \emptyset$ (nonempty, as α is algebraic). I is clearly an ideal, and $\mathbb{Q}[x]$ is a PID, so we may write $I = \langle f \rangle$ for some f . Since f is uniquely determined up to units, we may divide any choice of f by its leading coefficient to let f be monic. It remains to check that f is irreducible. Let $f = gh$; then $0 = f(\alpha) = g(\alpha)h(\alpha)$. It must be that one of $g(\alpha)$ or $h(\alpha)$ is zero, with the other being in I . Without loss of generality, let $g(x) \in I$; but then $f \mid g$, so $\deg g \geq \deg f$, whence h is a unit. $\square$

Day 3: Minimal Polynomials (Sep. 15, 2026)

Office hours will be held from 3–4pm on Wednesdays in BA6110. Last time, we defined the algebraic numbers, $\mathbb{Q}^{\text{alg}}$, and the algebraic integers, $\mathbb{Z}^{\text{alg}}$. We also saw that $\mathbb{Q}^{\text{alg}}$ is a field, and $\mathbb{Z}^{\text{alg}}$ is a ring.

Proposition 3.1. Let $\alpha \in \mathbb{Q}^{\text{alg}}$ be nonzero; there is a unique monic irreducible polynomial $f(x) \in \mathbb{Q}[x]$ such that $f(\alpha) = 0$. Moreover, if $g(x) \in \mathbb{Q}[x]$ such that $g(\alpha) = 0$, then $f \mid g$.

Definition 3.2. Let $\alpha \in \mathbb{Q}^{\text{alg}}$. Then the unique monic irreducible $f(x) \in \mathbb{Q}[x]$ with $f(\alpha) = 0$ is called the *minimal polynomial* of α . The *degree* of α is the degree of $f(x)$ itself.

If $\alpha \in \mathbb{Q}^{\text{alg}}$, write $\mathbb{Q}(\alpha) = \{h(\alpha) \mid h \in \mathbb{Q}[x]\}$. If $\alpha \in \mathbb{Z}^{\text{alg}}$, write $\mathbb{Z}[\alpha] = \{h(\alpha) \mid h \in \mathbb{Z}[x]\}$. Clearly, $\mathbb{Q}(\alpha) \subset \mathbb{C}$, and we call it the subring generated by $\mathbb{Q}$ and α in $\mathbb{C}$. Similarly, $\mathbb{Z}(\alpha)$ is the smallest subring of $\mathbb{C}$ containing α .

As an example, take $\mathbb{Q}(\sqrt{-5}) = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Q}\}$, and $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$.

Proposition 3.3. Let $\alpha \in \mathbb{Q}^{\text{alg}}$.

- (i) $\mathbb{Q}(\alpha) = \mathbb{Q}[x]/p(x)$, where p is the minimal polynomial of α .
- (ii) As a vector space over $\mathbb{Q}$, $\mathbb{Q}(\alpha)$ has basis $1, \alpha, \dots, \alpha^{n-1}$, where $n = \deg(\alpha)$.
- (iii) $\mathbb{Q}(\alpha)$ is a field.

Proof. Consider the map $\text{ev}_\alpha: \mathbb{Q}[x] \rightarrow \mathbb{Q}(\alpha)$, given by $h \mapsto h(\alpha)$. ev_α is a surjective ring homomorphism by definition of $\mathbb{Q}(\alpha)$ (cf. earlier), whence

$$\mathbb{Q}(\alpha) \cong \frac{\mathbb{Q}[x]}{\ker(\text{ev}_\alpha)}.$$

In particular, we saw that $\ker(\text{ev}_\alpha) = \{h \in \mathbb{Q}[x] \mid h(\alpha) = 0\} = \langle f(x) \rangle$, where $f(x) = a_0 + \dots + a_{n-1}x^{n-1} + x^n$, so we are done with (i). (ii) immediately follows from earlier. For (iii), pick $\beta \in \mathbb{Q}(\alpha)$ nonzero, and let $\beta = h(\alpha)$ with $h \in \mathbb{Q}[x]$, for which $f \nmid h$, since $h(\alpha) \neq 0$. By the division algorithm (or Bezout's lemma), there must exist polynomials $A, B \in \mathbb{Q}[x]$ such that $Af + Bh = 1$. Evaluating at α , we see that $A(\alpha)f(\alpha) = 0$, so $B(\alpha)h(\alpha) = 1$, so we have found an inverse $\beta^{-1} = B(\alpha) \in \mathbb{Q}(\alpha)$ as desired. $\square$

Example 3.4. We give some examples of irreducible polynomials.

- (i) The minimal polynomial of $\sqrt{-5}$ is $x^2 + 5$.³
- (ii) Let $\omega = \frac{1}{2}(-1 + \sqrt{-3}) = \exp(2\pi i/3)$. Clearly, $\omega^3 = 1$. Since $x^3 - 1$ is not irreducible, we see that we may factor and obtain the actual minimal polynomial for ω to be $x^2 + x + 1$.

Is $\frac{1}{2} \in \mathbb{Z}^{\text{alg}}$? Recall Gauss' lemma; suppose R is a UFD. A polynomial $f(x) \in R[x]$ is *primitive* if the GCD of its coefficients is 1; Gauss' lemma states that if $f, g \in R[x]$ is primitive, then fg is primitive as well. As a quick proof, suppose not; then there exists $p \in R$ prime such that $p \mid fg$; since R/p is an integral domain, let $\bar{f}, \bar{g} \in (R/p)[x]$ be the corresponding reductions. We have that $\bar{f} \cdot \bar{g} = 0$, but then it must be that $\bar{f} = 0$ or $\bar{g} = 0$, which is a contradiction.

³so... any idea why this is irreducible? "the eisenstein criterion?" *grimaces* wayyyy overkill...

Lemma 3.5. Let R be a UFD, with F its field of fractions. Suppose $g, h \in F[x]$ are monic such that $f := gh \in R[x]$. Then $g, h \in R[x]$.

Proof. Let c, d be the LCM of the denominators of the coefficients of g, h . Then cg and dh are primitive polynomials in $R[x]$. Gauss' lemma would then imply that $cg \cdot dh = (cd) \cdot gh$ is a primitive polynomial; but $gh \in R[x]$, so $cd \mid (cd) \cdot gh$ in $R[x]$, i.e., cd must be a unit in R . It follows that c and d are both units in R , whence $g, h \in R[x]$. $\square$

Corollary 3.6. If $\alpha \in \mathbb{Z}^{\text{alg}}$, then its minimal polynomial lies in $\mathbb{Z}[x]$.

Proof. Let $p(x) \in \mathbb{Q}[x]$ be the minimal polynomial of α . Since $\alpha \in \mathbb{Z}^{\text{alg}}$, there exists a monic $h(x) \in \mathbb{Z}[x]$ such that $h(\alpha) = 0$. $p(x) \mid h(x)$, i.e., $h(x) = p(x)q(x)$, for some q with rational coefficients. From here, the lemma tells us that $p, q \in \mathbb{Z}[x]$. $\square$

Corollary 3.7. $\mathbb{Q} \cap \mathbb{Z}^{\text{alg}} = \mathbb{Z}$.

Proof. Let $\alpha \in \mathbb{Q} \cap \mathbb{Z}^{\text{alg}}$. The minimal polynomial of α is $x - \alpha$, so $\alpha \in \mathbb{Z}$. $\square$

Definition 3.8. Algebraic numbers $\alpha, \beta \in \mathbb{Q}^{\text{alg}}$ are *conjugate* if they have the same minimal polynomial.

Lemma 3.9. We have some more properties of minimal polynomials.

- (i) If $f \in \mathbb{Q}[x]$ is monic and irreducible of degree d , then f has d distinct roots in $\mathbb{C}$.
- (ii) If α is an algebraic number of degree d , then there exist exactly d algebraic numbers conjugate to α .

Clearly, (i) implies (ii). We will prove this next lecture.

Day 4: Number Fields (Sep. 17, 2026)

Office hours have been moved to 2–3pm on Thursdays. Last time, given an algebraic number $\alpha \in \mathbb{Q}^{\text{alg}}$, we defined the conjugates of α to be the algebraic numbers with the same minimal polynomial, and we introduced the following lemma,

Lemma 4.1. (i) Let $f \in \mathbb{Q}[x]$ be monic and irreducible of degree d ; then f has d distinct roots in $\mathbb{C}$. (ii) If α is an algebraic number of degree d , then there exist exactly d algebraic numbers conjugate to α .

Proof. It is clear that (i) implies (ii), so it suffices to prove (i). First, $\mathbb{C}$ is algebraically closed, so $f = (x - \alpha_1) \dots (x - \alpha_d)$ with each $\alpha_i \in \mathbb{C}$. We need to show that $\alpha_i \neq \alpha_j$ for all i, j ; suppose $\alpha \in \mathbb{C}$ is a root of f with multiplicity greater than 1. Then f being monic and irreducible implies that f is the minimal polynomial of α . By taking the derivative of f , we see that $f'(\alpha) = 0$; but then $\deg f' = d - 1$, which is a contradiction. $\square$

Let's do some examples.

- (i) The conjugates of $\sqrt{5}$ are $\sqrt{5}$ and $\sqrt{-5}$, since $x^2 - 5 = 0$ is the minimal polynomial of $\sqrt{5}$. Recall that it is easy to check if a quadratic polynomial is irreducible (check the rational roots).
- (ii) Let $f(x) = x^4 - 10x^2 + 1$, which we know (from previous classes) may be factored as

$$f(x) = (x - \sqrt{2} - \sqrt{3})(x - \sqrt{2} + \sqrt{3})(x + \sqrt{2} - \sqrt{3})(x + \sqrt{2} + \sqrt{3}).$$

We claim that f is irreducible; there are many ways to check this, see the MAT347 notes. [In my opinion, the easiest would be to check the extensions of \$\mathbb{Q}\$.](#)

Corollary 4.2 (Viète). If $\alpha \in \mathbb{Q}^{\text{alg}}$ has minimal polynomial $f(x) = x^d + a_{d-1}x^{d-1} + \dots + a_0$ with conjugates $\alpha_1, \dots, \alpha_d$, then

$$\alpha_1 \dots \alpha_d = (-1)^d a_0, \quad \alpha_1 + \dots + \alpha_d = -a_{d-1}.$$

The proof is immediate; just write $f(x) = (x - \alpha_1) \dots (x - \alpha_d)$.

Definition 4.3. An *algebraic number field* is a subfield of F of $\mathbb{C}$ which is finite-dimensional over $\mathbb{Q}$.

Equivalently, an algebraic number field is a finite extension of $\mathbb{Q}$. (recall from MAT347 that any finite extension is generated by a single element, i.e., the primitive element theorem. cf. D&F §14.4.25). If F is a number field, then the *ring of integers* of F is denoted $\mathcal{O}_F = F \cap \mathbb{Z}^{\text{alg}}$.

Remark 4.4. We have the following properties of number fields.

- (i) Let $V = F \subset \mathbb{C}$ be a number field; then $\alpha \cdot V \subset V$ for $\alpha \in F$, and so $\alpha \in \mathbb{Q}^{\text{alg}}$. Specifically, any element of an algebraic number field is an algebraic number, $F \subset \mathbb{Q}^{\text{alg}}$.
- (ii) $\mathcal{O}_F$ is a subring of $\mathbb{C}$ containing $\mathbb{Z}$.
- (iii) $\mathbb{Q} \cap \mathbb{Z}^{\text{alg}} = \mathbb{Z}$, i.e., $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$.
- (iv) Let $\alpha \in \mathbb{Q}^{\text{alg}}$; we have that $\mathbb{Q}(\alpha) \subset \mathbb{C}$ is an algebraic number field. The primitive element theorem states that *every* number field is of this form.

Example 4.5. Let $d \in \mathbb{Z} \setminus \{0, 1\}$ be squarefree. Let $F = \mathbb{Q}(\sqrt{d}) = \{a + b\sqrt{d} \mid a, b \in \mathbb{Q}\}$. What is $\mathcal{O}_F$? Observe that

$$(a + b\sqrt{d})(a - b\sqrt{d}) = a^2 - db^2, \quad (a + b\sqrt{d}) + (a - b\sqrt{d}) = 2a,$$

so the minimal polynomial is $x^2 - 2ax + (a^2 - db^2)$. We want that $2a \in \mathbb{Z}$ and $a^2 - db^2 \in \mathbb{Z}$. If $d \equiv 2, 3 \pmod{4}$, then a, b are actually integers (checked by bashing). If $d \equiv 1 \pmod{4}$, then a, b are both integers, or half-integers. Thus,

$$\mathcal{O}_F = \begin{cases} \mathbb{Z} + \mathbb{Z}\sqrt{d}, & d \equiv 2, 3 \pmod{4}, \\ \mathbb{Z} + \mathbb{Z}\left(\frac{-1+\sqrt{d}}{2}\right), & d \equiv 1 \pmod{4}. \end{cases}$$

Note that, for $\alpha \in \mathbb{Z}^{\text{alg}}$ and $F = \mathbb{Q}(\alpha)$, $\mathcal{O}_F$ can be bigger than $\mathbb{Z}[\alpha]$.

Lemma 4.6. (i) If $\alpha \in \mathbb{Q}^{\text{alg}}$, there exists a nonzero integer $m \in \mathbb{N}$ such that $m\alpha \in \mathbb{Z}^{\text{alg}}$.

(ii) If F is a number field, then $F = \text{Frac}(\mathcal{O}_F)$.

Proof. We start by proving that (i) implies (ii). Since $\mathbb{Z} \subset \mathcal{O}_F$, we have that $\alpha = \frac{m\alpha}{m} \in \text{Frac}(\mathcal{O}_F)$. *Note that he skips a lot of intuition here, but the point is that $\mathcal{O}_F$ contains the integers, so from (i), we may write α as a fraction in terms of elements of $\mathcal{O}_F$; since $\mathcal{O}_F = F \cap \mathbb{Z}^{\text{alg}}$, it is enough to have $\alpha \in \mathcal{O}_F$ to get (ii).*

To prove (i), let $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$ (with $a_i \in \mathbb{Q}$) be the minimal polynomial of α . Then there exists m such that $ma_i \in \mathbb{Z}$ for all i ; it follows that we may multiply $f(x)$ throughout by m^n to see that $m\alpha$ is a root of

$$x^n + ma_{n-1}x^{n-1} + m^2a_{n-2}x^{n-2} + \dots + m^na_0 \in \mathbb{Z}[x]. \quad \square$$

We do more examples now.

- (i) Consider the Gaussian integers $F = \mathbb{Q}(i)$, where $i = \sqrt{-1}$ and $-1 \equiv 3 \pmod{4}$. Then $\mathcal{O}_F = \mathbb{Z}[i]$, for which $\mathcal{O}_F$ is a Euclidean domain with Euclidean norm $N(\alpha) = \alpha\bar{\alpha}$.
- (ii) Let $\omega = \exp(2\pi i/3) \in \mathbb{C}$. We have that $F = \mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$. Since $-3 \equiv 1 \pmod{4}$, we have that $\mathcal{O}_F = \mathbb{Z} + \mathbb{Z}\omega = \mathbb{Z}[\omega] \neq \mathbb{Z}[\sqrt{-3}]$. We claim that this is a Euclidean domain, with $N(\alpha) = \alpha\bar{\alpha}$ the Euclidean norm.
- (iii) Let $F = \mathbb{Q}\sqrt{-5}$. $-5 \equiv 3 \pmod{4}$, so $\mathcal{O}_F = \mathbb{Z}[\sqrt{-5}]$. Consider the Euclidean norm $N(a + b\sqrt{-5}) = a^2 + 5b^2 \in \mathbb{Z}_{\geq 0}$.

Lemma 4.7. The only units in $\mathbb{Z}[\sqrt{-5}]$ are -1 and 1 . The elements $3, 7, 1 + 2\sqrt{-5}, 1 - 2\sqrt{-5}$ are all irreducible.

Proof. We claim that $\alpha \in \mathbb{Z}[\sqrt{-5}]$ is a unit if and only if $N(\alpha) = 1$. Suppose $\alpha\beta = 1$; then $N(\alpha\beta) = N(\alpha)N(\beta) = 1$. Since $N(\alpha) \in \mathbb{Z}_{\geq 0}$, if $N(\alpha) = 1$, then $\alpha\bar{\alpha} = 1$, meaning $\bar{\alpha} = \alpha^{-1}$. In this manner, it is enough to solve $a^2 + 5b^2 = 1$, from which we get $b = 0$ and $a = \pm 1$, whence the units are ± 1 .

We now check that our claimed elements are irreducible. Suppose 3 is not irreducible. Let us write $3 = \alpha\beta$, where α, β are non-units. Per above, we see that this must mean that their norms are not 1 ; so

$$9 = N(3) = N(\alpha)N(\beta) \implies N(\alpha) = N(\beta) = 3;$$

since $\alpha = a + b\sqrt{-5}$ with $N(\alpha) = a^2 + 5b^2 \neq 3$, we get a contradiction, and we are done. The process is analogous to check that 7 is irreducible, i.e., we arrive at a contradiction when we find that $a^2 + 5b^2 = 7$ has no integer solutions.

To check that $1 \pm 2\sqrt{-5}$ is irreducible (we'll only check the $1 + 2\sqrt{-5}$ case because the other follows similarly), suppose $1 + 2\sqrt{-5} = \alpha\beta$, where α, β are non-units. Since

$$21 = N(1 + 2\sqrt{-5}) = N(\alpha)N(\beta),$$

we must have $N(\alpha)$ is 3 or 7, which is not possible as we saw earlier. Thus, we are done. $\square$

The moral of this situation is that, to analyze Diophantine equations, we really just want to bash things out with factorization. In addition, norms are really useful for testing these properties as well!

We recall some important properties of field extensions (perhaps, from MAT347). Let L/F be a finite extension (i.e., $F \subset L$). We denote the degree of the extension as

$$\deg L/F = |L : F| := \dim_F L.$$

As an example, $|\mathbb{Q}(\alpha) : \mathbb{Q}| = \deg \alpha$. The tower law states that, if L/M and M/F are finite extensions, and $\omega_1, \dots, \omega_n$ is a basis for L/M , then

$$L = M\omega_1 \oplus \dots \oplus M\omega_n.$$

We also have

$$|L : F| = |L : M| |M : F|.$$

We have the following key example. Let F be a number field, and take $\alpha \in F$; then

$$|F : \mathbb{Q}| = |F : \mathbb{Q}(\alpha)| |\mathbb{Q}(\alpha) : \mathbb{Q}|,$$

for which the latter is simply just $\deg \alpha$. In particular, $\deg \alpha \mid |F : \mathbb{Q}|$. What can we do with this? Suppose $\alpha = \sqrt{2} + \sqrt{3}$. $\mathbb{Q}(\alpha)$ contains $\sqrt{2}$ and $\sqrt{3}$. Since

$$\sqrt{2} = \frac{\alpha^3 - 9\alpha}{-4},$$

we see that $\mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\alpha)$. By writing

$$\deg \alpha = |F : \mathbb{Q}| = |F : \mathbb{Q}(\sqrt{2})| |\mathbb{Q}(\sqrt{2}) : \mathbb{Q}| = 4,$$

where $|\mathbb{Q}(\sqrt{2}) : \mathbb{Q}| = 2$, we see that $|F : \mathbb{Q}(\sqrt{2})| = 2$.

We now discuss norm and trace.

Definition 4.8. Let L/F be a finite extension. The *norm* of α , denoted $N_{L/F}(\alpha)$, is the determinant of the F -linear map $\phi_\alpha = \bullet\alpha : L \rightarrow L$. Similarly, the trace of α is $\text{Tr}_{F/L}(\alpha) := \text{tr}(\phi_\alpha)$.

Let us look at some examples.

- (i) $\mathbb{C}/\mathbb{R}$ has basis $\{1, i\}$. The matrix for multiplication by $x + iy$ is

$$\begin{pmatrix} x & -y \\ y & x \end{pmatrix},$$

(which you may perhaps recognize as the Cauchy–Riemann equations). Then $N_{\mathbb{C}/\mathbb{R}}(x + iy) = x^2 + y^2$, and $\text{Tr}_{\mathbb{C}/\mathbb{R}}(x + iy) = 2x$.

- (ii) $\mathbb{Q}(i)/\mathbb{Q}$ follows the exact same process. The norm and trace of $a + bi$ is $a^2 + b^2$, $2a$ respectively, since the basis and matrix are the same.
- (iii) $\mathbb{Q}(\omega)/\mathbb{Q}$, where $\omega^2 + \omega + 1 = 0$ (and $\omega = \exp(2\pi i/3)$), has basis $1, \omega$. Multiplication by $a + b\omega$ in this basis is

$$\begin{pmatrix} a & -b \\ b & a - b \end{pmatrix},$$

so $N_{\mathbb{Q}(\omega)/\mathbb{Q}}(a + b\omega) = a^2 + b^2 - ab$, and $\text{Tr}_{\mathbb{Q}(\omega)/\mathbb{Q}}(a + b\omega) = 2a - b$. The norm is the product of the conjugates, and conjugate in this situation is the complex conjugate (really, this has to do with the fact that the degree is 2; not every degree 2 extension is like this, though).

Proposition 4.9. We give the main properties of the norm here. $N_{L/F}: L \rightarrow F$ has the following properties,

- (i) $N_{L/F}(\alpha_1 \alpha_2) = N_{L/F}(\alpha_1) N_{L/F}(\alpha_2)$;
- (ii) $N_{L/F}(a) = a^{|L:F|}$ if $a \in F$;
- (iii) $N_{L/F}(\alpha) = 0$ if and only if $\alpha = 0$;
- (iv) If M is intermediate between L and F , then

$$N_{L/F} = N_{M/F} \circ N_{L/M}.$$

Proof. Properties (i) to (iii) are clear, so we will check (iv) only. We will start with a proof sketch in the special case that $\alpha \in M$; let $\omega_1, \dots, \omega_n$ be a basis for L/M . Then $L = M\omega_1 + \dots + M\omega_n$, and α acts block-diagonally. Thus, $N_{L/F}(\alpha) = N_{M/F}(\alpha)^n = N_{M/F} \circ N_{L/M}(\alpha)$. *Perhaps, this will be continued later?* $\square$

Proposition 4.10. $\text{Tr}_{L/F}$ has the following properties,

- (i) $\text{Tr}_{L/F}(\alpha_1 + \alpha_2) = \text{Tr}_{L/F}(\alpha_1) + \text{Tr}_{L/F}(\alpha_2)$;
- (ii) $\text{Tr}_{L/F}(a\alpha) = a \text{Tr}_{L/F}(\alpha)$ for $a \in F$ and $\alpha \in L$;
- (iii) $\text{Tr}_{L/F}(a) = |L:F| a$ if $a \in F$;
- (iv) If M is an intermediate field between L and F , then

$$\text{Tr}_{L/F} = \text{Tr}_{M/F} \circ \text{Tr}_{L/M}.$$

Proof. The proof is the same. $\square$

Remark 4.11. Let L/F be Galois. Then

$$N_{L/F}(\alpha) = \prod_{\sigma \in \text{Gal}(L/F)} \sigma(\alpha), \quad \text{Tr}_{L/F}(\alpha) = \sum_{\sigma \in \text{Gal}(L/F)} \sigma(\alpha).$$

In general, sum and products embed L into F . *I didn't catch what he said here; he'll probably restate this next class, so, we'll see then.*

We will do this next time.